

Some Class
Random Examples

Your Name

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Chapter 1

1.1 Random Examples

Definition 1.1.1: Limit of Sequence in $\mathbb{R}$

Let $\{s_n\}$ be a sequence in $\mathbb{R}$. We say

$$\lim_{n \rightarrow \infty} s_n = s$$

where $s \in \mathbb{R}$ if $\forall$ real numbers $\epsilon > 0 \exists$ natural number N such that for $n > N$

$$s - \epsilon < s_n < s + \epsilon \text{ i.e. } |s - s_n| < \epsilon$$

Question 1

Is the set $x\text{-axis} \setminus \{\text{Origin}\}$ a closed set

Solution: We have to take its complement and check whether that set is a open set i.e. if it is a union of open balls

Note:-

We will do topology in Normed Linear Space (Mainly $\mathbb{R}^n$ and occasionally $\mathbb{C}^n$) using the language of Metric Space

Remark 1.1.1 Strategy

For closed-set questions, proving that the complement is open is often the cleanest route.

Claim 1.1.1 Topology

Topology is cool

Example 1.1.1 (Open Set and Close Set)

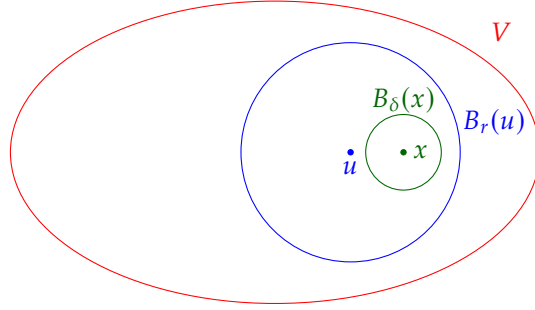
- Open Set:
- ϕ
 - $\bigcup_{x \in X} B_r(x)$ (Any $r > 0$ will do)
 - $B_r(x)$ is open
- Closed Set:
- $\overline{X}, \phi$
 - $\overline{B_r(x)}$
- $x\text{-axis} \cup y\text{-axis}$

Theorem 1.1 Open neighborhoods

If $x \in$ open set V then $\exists \delta > 0$ such that $B_\delta(x) \subset V$.

The reference [Theorem 1.1](#) is produced by `\Cref{th:open-neighborhoods}`.

Proof: By openness of V , $x \in B_r(u) \subset V$



Given $x \in B_r(u) \subset V$, we want $\delta > 0$ such that $x \in B_\delta(x) \subset B_r(u) \subset V$. Let $d = d(u, x)$. Choose δ such that $d + \delta < r$ (e.g. $\delta < \frac{r-d}{2}$). If $y \in B_\delta(x)$ we will be done by showing that $d(u, y) < r$ but

$$d(u, y) \leq d(u, x) + d(x, y) < d + \delta < r$$

☺

1.2 Reference Examples

Definition 1.2.1: Referenceable Definition

A labeled definition uses the third argument as the label key.

Lemma 1.2.1 Referenceable Lemma

A labeled lemma works the same way.

Theorem 1.2 Referenceable Theorem

A labeled theorem also works the same way.

[Definition 1.2.1](#), [Lemma 1.2.1](#), and [Theorem 1.2](#) are clickable references.

Reference Syntax

```
\begin{Definition}{Referenceable Definition}{referenceable-definition}
Body
\end{Definition}

\begin{Lemma}{Referenceable Lemma}{referenceable-lemma}
Body
\end{Lemma}

\begin{theorem}{Referenceable Theorem}{referenceable-theorem}
Body
\end{theorem}

\Cref{def:referenceable-definition}
\Cref{th:referenceable-lemma}
\Cref{th:referenceable-theorem}
```

Corollary 1.2.1

By the result of the proof, we can then show...

Lemma 1.2.2

Suppose $\vec{v}_1, \dots, \vec{v}_n \in \mathbb{R}^n$ is subspace of $\mathbb{R}^n$.

Proposition 1.2.1

$1 + 1 = 2$.

1.3 Random

Definition 1.3.1: Normed Linear Space and Norm $\|\cdot\|$

Let V be a vector space over $\mathbb{R}$ (or $\mathbb{C}$). A norm on V is function $\|\cdot\| : V \rightarrow \mathbb{R}_{\geq 0}$ satisfying

- ① $\|x\| = 0 \iff x = 0 \ \forall x \in V$
- ② $\|\lambda x\| = |\lambda| \|x\| \ \forall \lambda \in \mathbb{R}(\text{or } \mathbb{C}), x \in V$
- ③ $\|x + y\| \leq \|x\| + \|y\| \ \forall x, y \in V$ (Triangle Inequality/Subadditivity)

And V is called a normed linear space.

- Same definition works with V a vector space over $\mathbb{C}$ (again $\|\cdot\| \rightarrow \mathbb{R}_{\geq 0}$) where ② becomes $\|\lambda x\| = |\lambda| \|x\| \ \forall \lambda \in \mathbb{C}, x \in V$, where for $\lambda = a + ib$, $|\lambda| = \sqrt{a^2 + b^2}$

Example 1.3.1 (p -Norm)

$V = \mathbb{R}^m$, $p \in \mathbb{R}_{\geq 0}$. Define for $x = (x_1, x_2, \dots, x_m) \in \mathbb{R}^m$

$$\|x\|_p = \left(|x_1|^p + |x_2|^p + \dots + |x_m|^p \right)^{\frac{1}{p}}$$

(In school $p = 2$)

Special Case $p = 1$: $\|x\|_1 = |x_1| + |x_2| + \dots + |x_m|$ is clearly a norm by usual triangle inequality.

Special Case $p \rightarrow \infty$ ($\mathbb{R}^m$ with $\|\cdot\|_\infty$): $\|x\|_\infty = \max\{|x_1|, |x_2|, \dots, |x_m|\}$

For $m = 1$ these p -norms are nothing but $|x|$. Now exercise

Question 2

Prove that triangle inequality is true if $p \geq 1$ for p -norms. (What goes wrong for $p < 1$?)

Solution: For Property ③ for norm-2

When field is $\mathbb{R}$:

We have to show

$$\begin{aligned} \sum_i (x_i + y_i)^2 &\leq \left(\sqrt{\sum_i x_i^2} + \sqrt{\sum_i y_i^2} \right)^2 \\ \Rightarrow \sum_i (x_i^2 + 2x_i y_i + y_i^2) &\leq \sum_i x_i^2 + 2\sqrt{\left[\sum_i x_i^2 \right] \left[\sum_i y_i^2 \right]} + \sum_i y_i^2 \\ \Rightarrow \left[\sum_i x_i y_i \right]^2 &\leq \left[\sum_i x_i^2 \right] \left[\sum_i y_i^2 \right] \end{aligned}$$

So in other words prove $\langle x, y \rangle^2 \leq \langle x, x \rangle \langle y, y \rangle$ where

$$\langle x, y \rangle = \sum_i x_i y_i$$

Note:-

- $\|x\|^2 = \langle x, x \rangle$
- $\langle x, y \rangle = \langle y, x \rangle$
- $\langle \cdot, \cdot \rangle$ is $\mathbb{R}$ -linear in each slot i.e.

$$\langle rx + x', y \rangle = r\langle x, y \rangle + \langle x', y \rangle \text{ and similarly for second slot}$$

Here in $\langle x, y \rangle$ x is in first slot and y is in second slot.

Now the statement is just the Cauchy-Schwartz Inequality. For proof

$$\langle x, y \rangle^2 \leq \langle x, x \rangle \langle y, y \rangle$$

expand everything of $\langle x - \lambda y, x - \lambda y \rangle$ which is going to give a quadratic equation in variable λ

$$\begin{aligned} \langle x - \lambda y, x - \lambda y \rangle &= \langle x, x - \lambda y \rangle - \lambda \langle y, x - \lambda y \rangle \\ &= \langle x, x \rangle - \lambda \langle x, y \rangle - \lambda \langle y, x \rangle + \lambda^2 \langle y, y \rangle \\ &= \langle x, x \rangle - 2\lambda \langle x, y \rangle + \lambda^2 \langle y, y \rangle \end{aligned}$$

Now unless $x = \lambda y$ we have $\langle x - \lambda y, x - \lambda y \rangle > 0$ Hence the quadratic equation has no root therefore the discriminant is greater than zero.

When field is $\mathbb{C}$:

Modify the definition by

$$\langle x, y \rangle = \sum_i \bar{x}_i y_i$$

Then we still have $\langle x, x \rangle \geq 0$

1.4 Code Blocks

Newton's method for square roots

```
1  from __future__ import annotations
2
3
4  def sqrt_newton(a: float, *, tol: float = 1e-12, max_steps: int = 50) -> float:
5      """Approximate sqrt(a) by Newton iteration."""
6      if a < 0:
7          raise ValueError("a must be nonnegative")
8      if a == 0:
9          return 0.0
10
11     x = max(1.0, a)
12     for step in range(1, max_steps + 1):
13         next_x = 0.5 * (x + a / x)
14         error = abs(next_x - x)
15
16         if error <= tol * max(1.0, next_x):
17             print(f"converged after {step} steps")
18             return next_x
19
20     x = next_x
21
22     raise RuntimeError("Newton iteration did not converge")
23
24
25 print(sqrt_newton(2.0))
```

1.5 Algorithms

Algorithm 1: Projected gradient descent

Input: Objective f , feasible set C , initial point x_0 , step sizes η_t

Output: Approximate minimizer x_t

```
1  for  $t = 0, 1, \dots, T - 1$  do
2       $g_t \leftarrow \nabla f(x_t);$                                      // gradient at the current iterate
3       $y_{t+1} \leftarrow x_t - \eta_t g_t;$ 
4       $x_{t+1} \leftarrow \Pi_C(y_{t+1});$                              // project back onto C
5      if  $\|x_{t+1} - x_t\| < \varepsilon$  then
6          return  $x_{t+1};$ 
7  return  $x_T;$ 
```

1.6 Syntax Reference

Short Commands

```
\dfn{Title}{Body}
\dfnc{Title}{Body}
\rmk{Title}{Body}
\qs{Title}{Body}
\nt{Body}

\thm{Title}{Body}
\cor{Title}{Body}
\mma{Title}{Body}
\mer{Title}{Body}
\mprop{Title}{Body}
\clm{Title}{label-name}{Body}
\wc{Title}{Body}

\pf{Proof name}{Body}
\sol Body
\begin{sol}
Body
\end{sol}
\solve{Body}
```

Theorem Environments

```
\begin{theorem}{Title}{label-name}
Body
\end{theorem}

\begin{Definition}{Title}{label-name}
Body
\end{Definition}

\begin{Remark}{Title}{label-name}
Body
\end{Remark}

\begin{question}{Title}{label-name}
Body
\end{question}

\begin{solution}
Body
\end{solution}
```

Code Helpers

```
\codeinline|inline code|
\codefile{path/to/file.py}{Python}

\begin{codeblock}[title={Optional title}]{Python}
print("hello")
\end{codeblock}

\begin{consoleblock}
python script.py
\end{consoleblock}
```

LaTeX Code Block Syntax

```
\begin{latexcodeblock}[title={Optional title}]  
\dfn{Title}{Body}  
\end{latexcodeblock}
```

Algorithm Syntax

```
\begin{algorithm}[H]  
\caption{Algorithm title}  
\KwIn{Input data}  
\KwOut{Output data}  
\For{$t = 0, \ldots, T-1$}{  
  do something\  
  \tcp*{inline comment}  
}  
\Return result\  
\end{algorithm}
```